REVISION 2026

Basic Surveying Lab

Diploma in Civil Engineering

Code: 2019 | Credits: 1.5 | Semester: 2

Course Outcomes (CO)

- **CO1:** Perform linear and angular measurements using chain and compass. (Apply)
- **CO2:** Determine level differences using simple and differential levelling. (Apply)
- **CO3:** Plot longitudinal profiles and prepare contour maps. (Apply)
- **CO4:** Measure horizontal angles and bearings using a theodolite. (Apply)

Module 1: Chain & Compass Survey

Expt 1: Equipment Introduction

Demonstration of basic survey equipment: Chain, Cross-staff, Plane table, and Compass.

Expt 3: Compass Survey

Determine inaccessible distances between points using compass and tape measurements.

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Module 2: Levelling

Expt 4: Simple Levelling

Determine level difference between two points using a Dumpy level.

Expt 5: Rise and Fall Method

Determine reduced levels (RL) by the Rise and Fall method for a series of points.

(RL) .

Module 3: Profile & Contouring

Expt 7: Profile Levelling

Plot longitudinal profiles and cross sections for the alignment of a road.

Expt 9: Contouring

Take spot levels and prepare a contour map for a given area.

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Module 4: Theodolite Survey

Expt 10: Repetition Method

Measure horizontal angles with high precision using the repetition method with a theodolite.

Expt 12: Prolonging a Line

Prolong a straight line and measure the bearing of a line using a theodolite.

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Practice Model Question Paper

Practice Model Paper — not an official REV2026 paper

1. Determine the level difference between two given points using the Height of Instrument (HI) method.

2. Measure the horizontal angle between two objects using the repetition method with a theodolite.

Source Declaration

Curriculum Scheme: Revision 2026

Subject Name: Basic Surveying Lab

Subject Code: 2019

Official Source URL: SITTTR Link

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